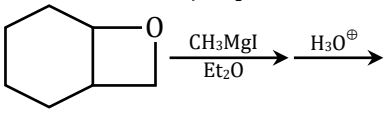
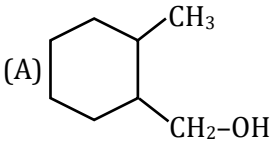
RACE
Regular Analysis Through Continuous Exercise

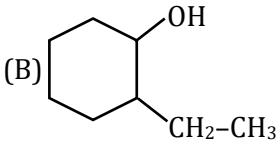
Grignard's Reagent : RACE-01

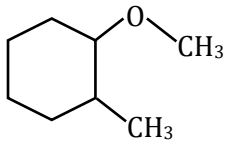
Reaction of Grignard Reagent

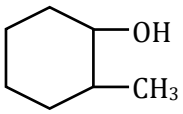
1. What is the major product of the following reaction?



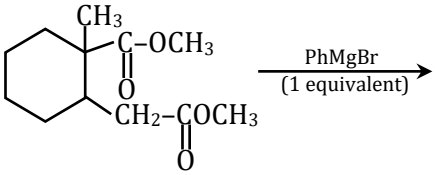
(A) 

(B) 

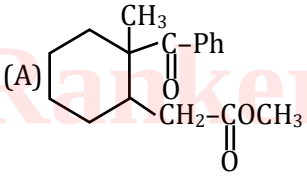
(C) 

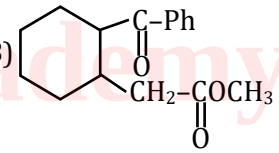
(D) 

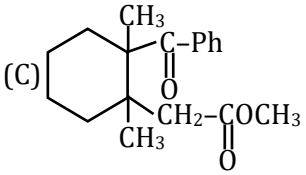
CGR113

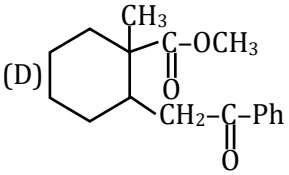
2. 

Product (P) is :

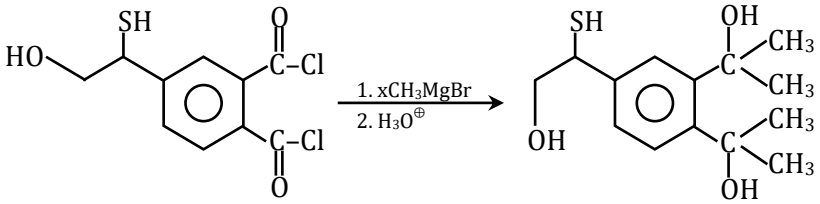
(A) 

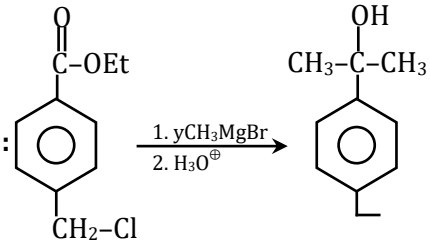
(B) 

(C) 

(D) 

CGR114

3. **Reaction-1 :** 

Reaction-2: 

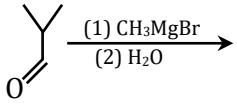
What is the ratio (x/y) is above problem?

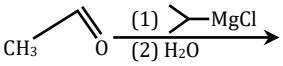
(A) 1.5 (B) 2 (C) 2.5 (D) 3

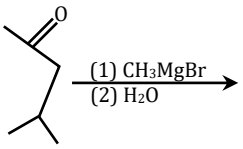
CGR115

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4. In which of the following reaction 2° alcohol is obtained as a product?

(A) 

(B) 

(C) 

(D) Both (A) and (B)

CGR116

5. Identify product Z in the following reaction sequence :

$$H_2C=CHCH_2Br \xrightarrow{NaCN} Y \xrightarrow[2.H_3O^+]{1.C_6H_5MgBr,\ diethylether} Z$$

(A) $H_2C = CHCH_2\overset{\overset{O}{\|}}{C}C_6H_5$

(B) $H_2C = CHCH_2NH\overset{\overset{O}{\|}}{C}C_6H_5$

(C) $H_2C = CHCH_2\underset{\underset{OH}{|}}{C}HC_6H_5$

(D) $H_2C = CHCH_2\underset{\underset{NH_2}{|}}{C}HC_6H_5$

CGR117

6. $CH_3 - CH = CH_2 \xrightarrow[hv\ (low\ conc.)]{Br_2} \xrightarrow[Dry\ ether]{Mg} \xrightarrow[NH_4Cl]{CH_3-\overset{\overset{O}{\|}}{C}-CH_3} \xrightarrow[\Delta]{H^+}$ (X) (major)

End product (X) of the above reaction is :

(A) $CH_2 = CH - CH_2 - \overset{\overset{CH_2}{\|}}{C} - CH_3$

(B) $H_2C = CH - CH = \underset{\underset{CH_3}{|}}{C} - CH_3$

(C) $H_2C = CH - CH_2 - \underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}} - CH_3$

(D) $H_2C = CH - CH_2 - \underset{\underset{CH_3}{|}}{CH} - CH_2 - OH$

CGR118

7. $\begin{matrix} CH_2—OH \\ | \\ CH_2—OH \\ | \\ CH_2—SH \end{matrix} + \underset{(excess)}{CH_3MgBr} \rightarrow xCH_4$

What is the value of x in the above reaction?>

(A) 1 (B) 2 (C) 3 (D) 4

CGR119

8. 0.40 g of an organic compound (A), (M.F.-C_5H_8O) reacts with x mole of CH_3MgBr to liberate 224 mL of a gas at STP. With excess of H_2, (A) gives pentan-1-ol. The correct structure of (A) is :

(A) $CH_3-C\equiv C-CH_2-CH_2-OH$

(B) $CH_3-CH_2-C\equiv C-CH_2-OH$

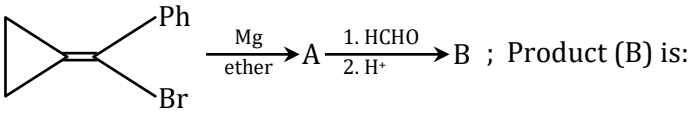
(C) $H-C\equiv C-CH_2-CH_2-CH_2OH$

(D) $H-C\equiv C-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

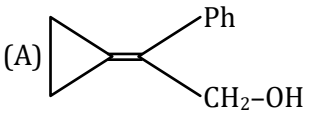
CGR120

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9. 

; Product (B) is:

(A) 

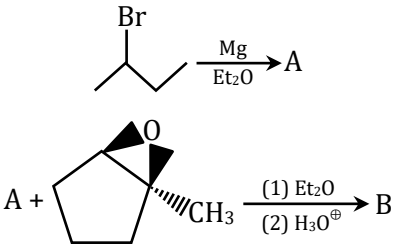
(B) Ph–C≡C–CH_2–CH_2–CH_2–OH

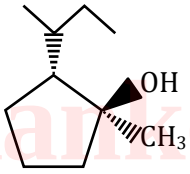
(C) Ph–C≡C–CH_2–CH_2–OH

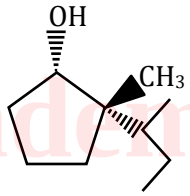
(D) Ph–CH_2–C≡C–CH_2–CH_2–OH

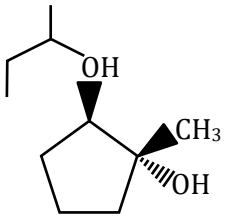
CGR121

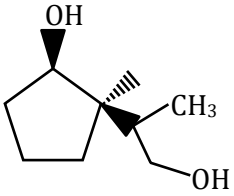
10. What is the product (B) of the following reaction sequence?



(A) 

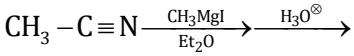
(B) 

(C) 

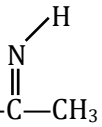
(D) 

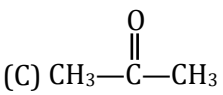
CGR122

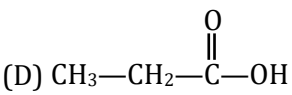
11. What is the major product of the following reaction?



(A) $CH_3 - CH_2 - NH - CH_3$

(B) 

(C) 

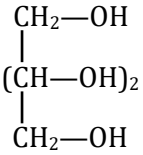
(D) 

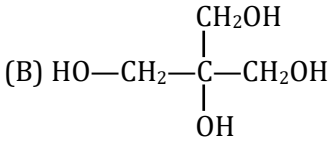
CGR123

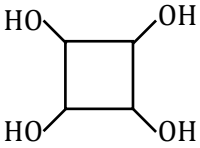
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12. What 20 g of a compound (A) (M.F. = $C_4H_{10}O_4$) reacts with excess of of CH_3 MgBr, 14.6 L of CH_4 is obtained at STP. What is structureal formula of (A)?

(A) 

(B) 

(C) 

(D) Both (A) & (B)

CGR124

13. How many isomers of $C_4H_{10}O$ reacts with CH_3MgBr to evolve CH_4 gas? (Excluding stereoisomer)

CGR125

14. How many carbonyl isomers of $C_5H_{10}O$ which reacts with PhMgBr to give racemic mixtures?

CGR126

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	D	A	B	C	D	B	A
Question	11	12	13	14						
Answer	C	D	4	6						

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